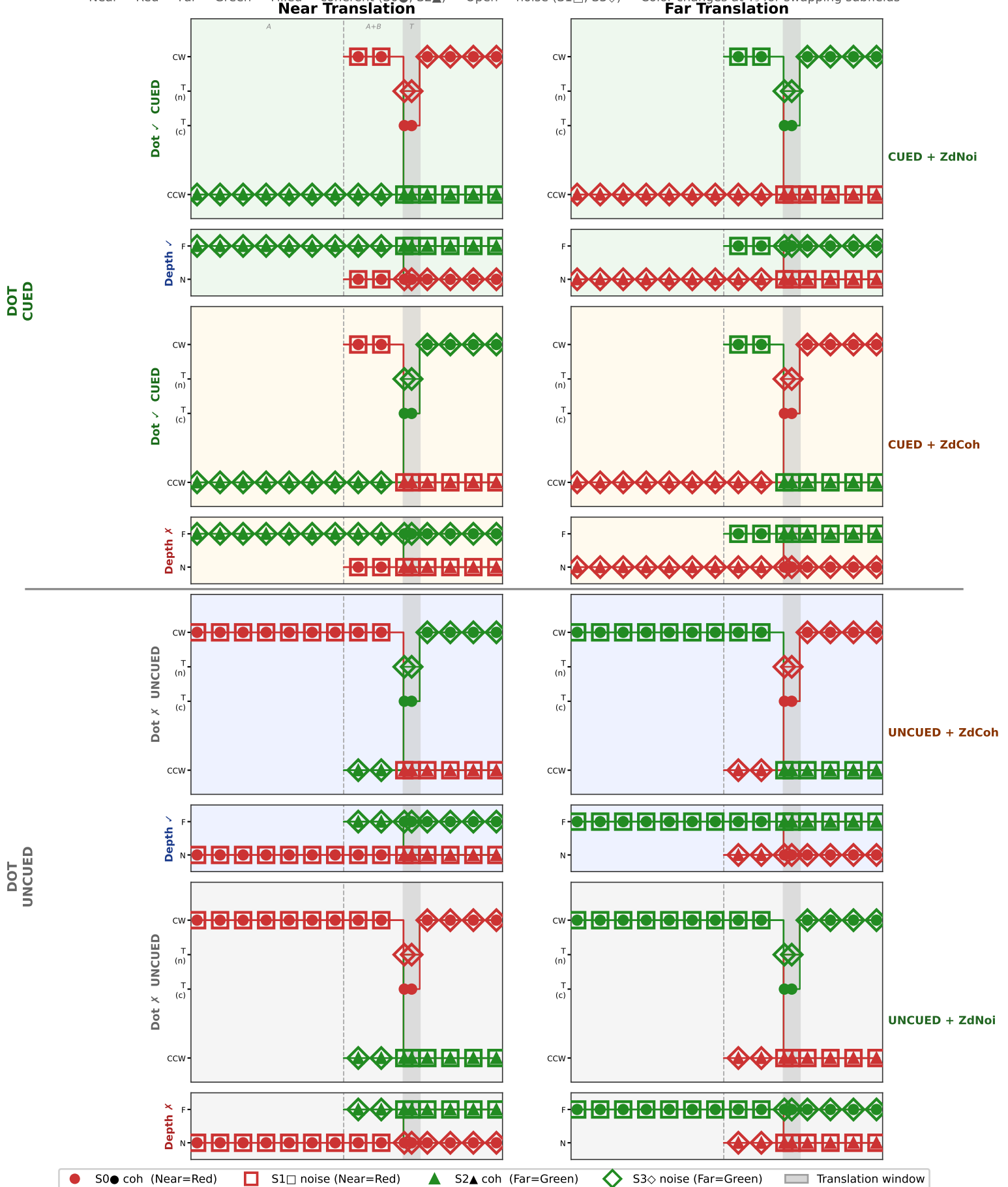


Depth + Color Design — ZdCoh / ZdNoi

Factor 1 — Dot Cueing (✓/✗): does the delayed-onset dot field translate?

Factor 2 — Depth Cueing (✓/✗): does translation occur at the depth where the cued field appeared?

Near = Red · Far = Green · Filled = coherent (S0●, S2▲) · Open = noise (S1□, S3◇) · Color changes at T_0 for swapping subfields



Depth + Color Design — Permutations & Analysis Framework

Trial-type permutations (32 = 2 swap types × 4 binary variables × 8 TransDeg; N baseline dropped — see prior experiments)

#	Swap	Cond	DFD	NClr	Rc	TrDep	TrClr	TrRot	CuClr	IdPrsv	F1 / F2 / F3 / F4 / F5 / F6
1	ZdCoh	CUED	N	R	0	N	R	CCW	R	NO	CU N R CCW R N
2	ZdCoh	CUED	N	R	1	N	R	CW	R	NO	CU N R CW R N
3	ZdCoh	CUED	N	G	0	N	G	CCW	G	NO	CU N G CCW G N
4	ZdCoh	CUED	N	G	1	N	G	CW	G	NO	CU N G CW G N
5	ZdCoh	CUED	F	R	0	F	G	CCW	G	NO	CU F G CCW G N
6	ZdCoh	CUED	F	R	1	F	G	CW	G	NO	CU F G CW G N
7	ZdCoh	CUED	F	G	0	F	R	CCW	R	NO	CU F R CCW R N
8	ZdCoh	CUED	F	G	1	F	R	CW	R	NO	CU F R CW R N
9	ZdCoh	UNCUED	N	R	0	F	G	CW	R	NO	UN F G CW R N
10	ZdCoh	UNCUED	N	R	1	F	G	CCW	R	NO	UN F G CCW R N
11	ZdCoh	UNCUED	N	G	0	F	R	CW	G	NO	UN F R CW G N
12	ZdCoh	UNCUED	N	G	1	F	R	CCW	G	NO	UN F R CCW G N
13	ZdCoh	UNCUED	F	R	0	N	R	CW	G	NO	UN N R CW G N
14	ZdCoh	UNCUED	F	R	1	N	R	CCW	G	NO	UN N R CCW G N
15	ZdCoh	UNCUED	F	G	0	N	G	CW	R	NO	UN N G CW R N
16	ZdCoh	UNCUED	F	G	1	N	G	CCW	R	NO	UN N G CCW R N
17	ZdNoi	CUED	N	R	0	N	R	CCW	R	YES	CU N R CCW R Y
18	ZdNoi	CUED	N	R	1	N	R	CW	R	YES	CU N R CW R Y
19	ZdNoi	CUED	N	G	0	N	G	CCW	G	YES	CU N G CCW G Y
20	ZdNoi	CUED	N	G	1	N	G	CW	G	YES	CU N G CW G Y
21	ZdNoi	CUED	F	R	0	F	G	CCW	G	YES	CU F G CCW G Y
22	ZdNoi	CUED	F	R	1	F	G	CW	G	YES	CU F G CW G Y
23	ZdNoi	CUED	F	G	0	F	R	CCW	R	YES	CU F R CCW R Y
24	ZdNoi	CUED	F	G	1	F	R	CW	R	YES	CU F R CW R Y
25	ZdNoi	UNCUED	N	R	0	F	G	CW	R	YES	UN F G CW R Y
26	ZdNoi	UNCUED	N	R	1	F	G	CCW	R	YES	UN F G CCW R Y
27	ZdNoi	UNCUED	N	G	0	F	R	CW	G	YES	UN F R CW G Y
28	ZdNoi	UNCUED	N	G	1	F	R	CCW	G	YES	UN F R CCW G Y
29	ZdNoi	UNCUED	F	R	0	N	R	CW	G	YES	UN N R CW G Y
30	ZdNoi	UNCUED	F	R	1	N	R	CCW	G	YES	UN N R CCW G Y
31	ZdNoi	UNCUED	F	G	0	N	G	CW	R	YES	UN N G CW R Y
32	ZdNoi	UNCUED	F	G	1	N	G	CCW	R	YES	UN N G CCW R Y

Depth + Color Design — Analysis Framework

Abbreviations

Design variables (trial parameters)		Derived analysis variables	
Swap Cond	ZdCoh / ZdNoi — which subfields swap depth+color at T _s CUED: delayed-onset field translates UNCUEd: it does not	TrDep	Translator Depth — depth plane of the translating field
DFD	Delayed Field Depth — depth plane of the onset-cued field (N=Near, F=Far)	TrClr	Translator Color — pre-swap color of translating field
NClr	NearColor — color assigned to Near plane (R=Red, G=Green; Far gets the other)	TrRot	Translator Rotation direction (CW / CCW)
Rc	RotConfig — 0: delayed field rotates CCW 1: delayed field rotates CW	CuClr	Cued Color — color of the delayed (onset-cued) field
		IdPrsv	Identity Preserved — YES (ZdNoi): translator keeps depth+color after T _s ; NO (ZdCoh): it changes
		Subfields (4 per trial)	
		S0●	Coherent CW — the translating field in CUED trials
		S1□	Noise CW
		S2▲	Coherent CCW — the translating field in UNCUEd trials
		S3◇	Noise CCW
		Swap conditions	
		ZdCoh	Coherent subfields S0+S2 swap depth+color at T _s ; translator (S0 in CUED) changes identity
		ZdNoi	Noise subfields S1+S3 swap depth+color at T _s ; translator (S0 in CUED) retains its depth+color identity

Factor Summary

F1 Dot-field cueing [CUED vs UNCUEd]

Temporal-onset cueing effect. The primary question: does the delayed onset advantage survive the depth+color swap?

F2 Translator depth [Near vs Far (= TransDepth)]

Is Near or Far translation easier? Orthogonal to F3 (each depth is equally often R or G via NearColor counterbalancing). Depth and color ARE separable within this experiment.

F3 Translator color [Red vs Green (pre-swap)]

Does the pre-swap color of the translating field affect performance? Directly tests the red/green asymmetry hypothesis from DepthSwapCtrl.

F4 Translator rotation [CW vs CCW]

Expected null. Included for completeness.

F5 Cued-field color [R vs G (= color of delayed field)]

In CUED: equals F3 (same field). In UNCUEd: diverges from F3 — cued field is not the translator. Tests whether the observer's color-based attention bias affects UNCUEd performance.

F6 Field-identity cueing [ZdNoi vs ZdCoh (IdPrsv = Y vs N)]

The new factor enabled by including ZdNoi. ZdNoi preserves the cued translator's depth+color; ZdCoh disrupts it. The ZdNoi – ZdCoh cueing difference = benefit of stable depth+color identity for the cued translator. This is the depth+color FIELD-IDENTITY cueing effect.

Key Within-Experiment Comparisons

N vs ZdCoh

Total (depth+color) identity contribution to cueing. ZdCoh removes BOTH depth AND color as post-T_s field-identity cues. Any cueing that survives in ZdCoh is pure temporal-onset (dot) cueing.

N vs ZdNoi

Cost of disrupting the NOISE group's depth+color identity, while the translator itself remains in its original plane and color. If ZdNoi ≈ N, noise-group stability does not matter. If ZdNoi < N, even peripheral disruption hurts tracking.

ZdCoh vs ZdNoi (= F6)

The critical dissociation: matched disruption count, different which subfields. ZdNoi – ZdCoh = benefit of keeping the TRANSLATOR in a stable depth+color identity. Directly answers: is depth+color field-identity cueing driven by the translator specifically, or by the group as a whole?

F1 × F6 (Cuing × IdPrsv)

Is the CUED>UNCUEd advantage larger when translator identity is preserved (ZdNoi) than disrupted (ZdCoh)? This interaction is the core measure of depth+color field-identity cueing.

Cross-experiment: ZdCoh+color vs ZdA+all-red (existing)

Isolates the COLOR contribution specifically: same disruption pattern, different whether color also changes. Requires the current data vs prior DepthSwapCtrl.